MONTH 1

Month 1: Front-End Development

Week 1: HTML5 & CSS3 Fundamentals

- Tasks:
  - Learn the basics of HTML5 and CSS3.
  - Explore semantic HTML and modern CSS features.
  - Project: Build a simple landing page for a small business or service (e.g., a local restaurant or a personal portfolio).
- DC Info Tech Real-World Application: Interns will create basic web pages for client landing pages or in-house demo projects.

Week 2: Responsive Web Design

- Tasks:
  - Understand CSS Flexbox and Grid systems for responsive layouts.
  - Implement media queries for mobile, tablet, and desktop views.
  - Project: Refactor the landing page into a responsive design that works across devices.
- DC Info Tech Real-World Application: Convert existing web templates used by DC Info Tech for clients into responsive designs.

Week 3: Introduction to JavaScript

- Tasks:
  - Learn JavaScript basics, including DOM manipulation and event handling.
  - Project: Add interactivity to the landing page (e.g., form validation, interactive elements like sliders or modals).
- DC Info Tech Real-World Application: Interns will be tasked with adding dynamic features to internal websites or client pages, such as interactive feedback forms.

MONTH 2

Week 4: JavaScript Framework (React.js)

- Tasks:
  - Learn the basics of React.js, including components, state, and props.
  - Project: Develop a small dynamic component like a blog post list or a product gallery using React.
- DC Info Tech Real-World Application: Interns will assist with front-end components for ongoing client projects using React.js, such as developing new features for client dashboards.

Month 2: Back-End Development & APIs

Week 5: Introduction to Node.js & Express

- Tasks:
  - Set up a Node.js and Express server.
  - Understand RESTful API principles.
  - Project: Create a basic RESTful API for a simple service like a task manager.
- DC Info Tech Real-World Application: Interns will work on small API endpoints for internal tools or client applications.

Week 6: Database Integration (MongoDB)

- Tasks:
  - Learn MongoDB basics and integrate it with Node.js using Mongoose ORM.
  - Implement CRUD operations.
  - Project: Add MongoDB support to the API created in Week 5 (e.g., storing and retrieving tasks).
- DC Info Tech Real-World Application: Integrating MongoDB into internal project management systems or client solutions.

Week 7: Authentication & Authorization

- Tasks:
  - Implement JWT-based authentication for user login and role-based access.
  - Project: Secure the task manager API with login and user roles.
- DC Info Tech Real-World Application: Work on adding authentication to client or internal systems such as employee or client dashboards.

MONTH 3

Week 8: Full-Stack Project

- Tasks:
  - Build a full-stack web application combining React.js (frontend) and Node.js/MongoDB (backend).
  - Project: Create a full-fledged web application, such as an admin panel or small e-commerce site.
- DC Info Tech Real-World Application: Develop a working prototype for a real-world DC Info Tech client or internal project.

Month 3: Advanced Topics, Collaboration, and Deployment

Week 9: Version Control & Collaboration

- Tasks:
  - Learn Git for version control and work with GitHub repositories.
  - Collaborate on code using branching and pull requests.
  - Project: Work with another intern to complete a small feature for a web project.
- DC Info Tech Real-World Application: Work with DC Info Tech developers, contributing to real client projects or internal tools, following Git workflows.

Week 10: Deployment & Hosting

- Tasks:
  - Learn how to deploy web applications using services like Netlify or Heroku.
  - Project: Deploy the full-stack web app built in Week 8 to a live server and configure a custom domain.
- DC Info Tech Real-World Application: Interns will help deploy a client or internal web application to production, ensuring it is publicly accessible.

Week 11: Performance Optimization

- Tasks:
  - Learn web performance optimization techniques such as caching, minification, and lazy loading.
  - Project: Optimize the deployed web app, improving load times and user experience.
- DC Info Tech Real-World Application: Apply performance optimization techniques to DC Info Tech's ongoing projects or revamp client websites for faster load times.

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WEEK 12: FINAL FULL-STACK PROJECT

TASKS:

DEVELOP A FINAL FULL-STACK PROJECT FROM SCRATCH BASED ON REAL-WORLD SCENARIOS.

PROJECT: CHOOSE FROM EXAMPLES SUCH AS:

A CLIENT DASHBOARD WITH USER MANAGEMENT AND DATA VISUALIZATION.

A TASK MANAGEMENT SYSTEM FOR DC INFO TECH TEAM MEMBERS.

A SMALL E-COMMERCE PLATFORM WITH CART AND CHECKOUT FEATURES.

DC INFO TECH REAL-WORLD APPLICATION: PRESENT AND DEPLOY THE FINAL PROJECT, SHOWCASING IT AS A LIVE SOLUTION, AND CONTRIBUTE TO AN ONGOING DC INFO TECH CLIENT SOLUTION.

. INCLUDE DOCUMENTATION THAT OUTLINES THE PROJECT'S FUNCTIONALITY, ARCHITECTURE, AND HOW TO DEPLOY IT USE NETLIFY OR GITHUB LINK IS ALSO OK AFTER DONE PLEASE FILL UP THE FORM

INTERNSHIP COMPLETION FORM